# FreqODES: Frequency-Domain Guided Online Domain Adaptation with Expert-in-the-Loop Active Learning

Replication and extension of ODES (MICCAI 2025)

Group A12

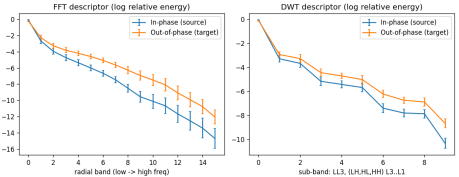
Vishnu Vardhan (CB.SC.U4AIE23043) Nithya Sai (CB.SC.U4AIE23021)

22AIE437 Medical Image Processing
B.Tech CSE (AI), Amrita School of Artificial Intelligence

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# Problem and motivation

- Abdominal organ segmentation (liver, kidneys, spleen) from MRI supports surgical planning, volumetry and follow-up.
- A network trained on one scanner / sequence (*source*) degrades on another (*target*): **domain shift**.
- Example: CHAOS T1-DUAL **in-phase** $\rightarrow$ **out-of-phase**: same patients, same organs, yet Dice of our source model drops from training level to **54.97%**.
- Retraining per hospital needs costly expert labels; unsupervised test-time adaptation relies on noisy pseudo-labels.
- **Online active domain adaptation**: adapt patient-by-patient, asking an expert to label only a tiny budget of pixels.




Same anatomy, different MR sequence: OOP has more high-frequency energy (FFT bands, left; DWT sub-bands, right).

# Dataset, task and objectives

**Dataset: CHAOS challenge** [2] (MR, T1-DUAL)

- 20 patients, 647 axial slices (26–50 per patient), $256 \times 256$ / $288 \times 288$, resized to $256^2$
- Labels: liver, right kidney, left kidney, spleen (+ background) = 5 classes
- **Source**: in-phase (IP), all 20 volumes, used for training
- **Target**: out-of-phase (OOP), the 20 volumes arrive as an *online stream*, one patient = one batch (paper protocol, no train/test split on target)

**Task**: 2-D multi-class **segmentation**

**Objectives**

1. Reproduce ODES [1] on CHAOS IP→OOP.
2. Replace its BN-statistics image-pruning score with an interpretable **FFT / DWT frequency-domain** shift score (FreqODES).
3. Compare at equal annotation budget ($K = 10, 50, 100\%$ slices, $b = 1\%$ pixels).

**Metrics**: Dice (DSC), Jaccard (IoU) per organ and mean; adaptation time; paired Wilcoxon test over the 20 patients.

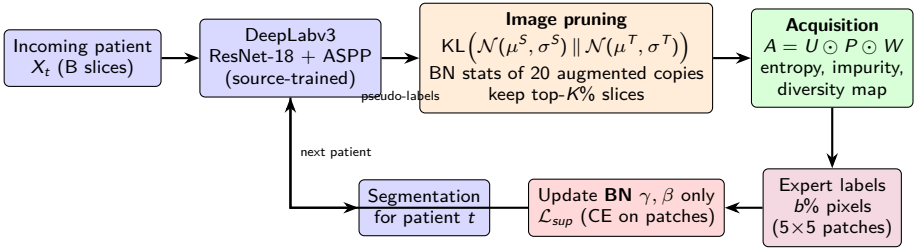
# Base paper

## ODES: Online Domain Adaptation with Expert Guidance for Medical Image Segmentation

M. S. Islam, S. Nag, A. Dutta, S. M. Ahmed, F. F. Niloy, S. Bera, A. K. Roy-Chowdhury
University of California, Riverside — **MICCAI 2025** [1]. Code: `github.com/ShazidAraf/ODES`

- First active-learning based *online* UDA framework for medical segmentation: source-free, storage-free, one pass per batch.
- Four stages per incoming patient: **infer** → **prune images** → **acquire patch labels** → **update BN layers**.
- Contributions: (i) **image pruning** by BN-statistics KL divergence; (ii) **diversity-aware acquisition** (entropy × regional impurity × farthest-first diversity map).
- Evaluated on CHAOS IP→OOP, CHAOS→DUKE, BMC→RUNMC; beats TENT, CoTTA, SaTTCA, ...

# ODES architecture




- **Backbone**: DeepLabv3 [3], ResNet-18 encoder (ImageNet init), ASPP, low-level skip, 5-class head.
- **Pruning**: each slice is augmented (flip / rotation), passed through a copy of the net; BN statistics of the last 9 BN layers are compared with source running statistics (Eq. 1 of [1]).
- **Acquisition**: $U$ = patch-mean entropy, $P$ = regional impurity [4], $W$ = Gaussian map around farthest-first-traversal picks (spatial $\times$ feature).

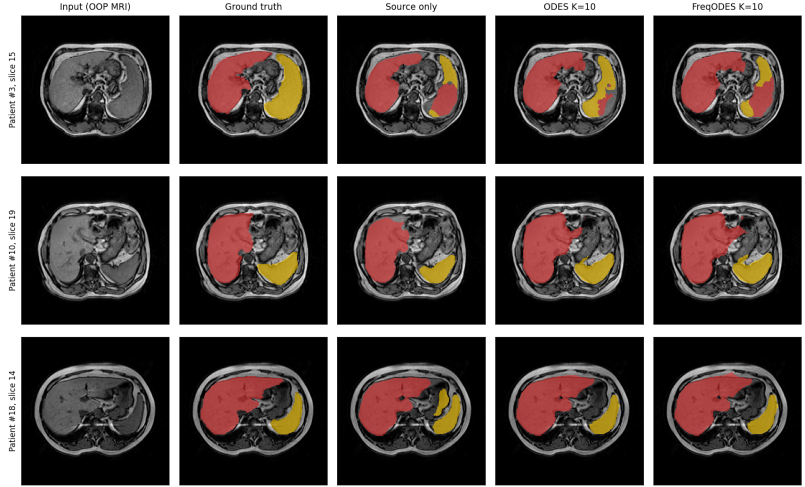
# Replicated vs. reported results (CHAOS IP $\rightarrow$ OOP, $b = 1\%$)

Table 1: Dice (%) per organ; Jaccard (IoU, %) mean. Paper values from Table 1 of [1].

| | Setting | Liver | L. Kidney | R. Kidney | Spleen | **Mean DSC** | **Mean IoU** |
|---|---|---|---|---|---|---|---|
| Source only | Paper | 87.77 | 37.97 | 18.92 | 67.31 | 52.99 | – |
| | Ours | 85.89 | 29.84 | 33.74 | 70.43 | 54.97 | 42.74 |
| ODES $K$=100 | Paper | 88.90 | 72.41 | 74.55 | 78.56 | 78.61 | – |
| | Ours | 87.89 | 76.90 | 75.28 | 78.00 | **79.52** | 66.96 |
| ODES $K$=50 | Paper | 89.13 | 72.30 | 74.43 | 78.36 | 78.56 | – |
| | Ours | 88.02 | 76.66 | 75.71 | 77.89 | **79.57** | 67.00 |
| ODES $K$=10 | Paper | 88.41 | 71.45 | 70.22 | 77.12 | 76.80 | – |
| | Ours | 87.88 | 75.19 | 74.18 | 78.06 | **78.83** | 66.24 |

- Source-only and adapted results are within $\approx$2 DSC of the paper; the adaptation gain (54.97$\rightarrow$79.52, $\approx +25$) and the $K$ trend ($100 \approx 50 > 10$) are reproduced.
- No source checkpoint or training recipe is released, so we trained our own source model; organ-level differences (e.g. kidneys) come from that.
- Paper reports no Jaccard; ours is measured per volume.

# Qualitative results: input, ground truth, predicted masks




Red: liver, green: right kidney, blue: left kidney, yellow: spleen. Three target (OOP) patients, slice with largest organ area.

# Code structure and training settings

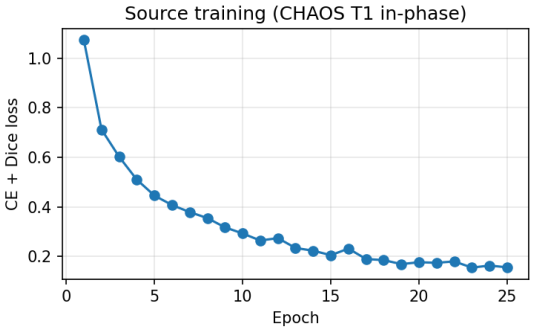
```
prepare_chaos.py    DICOM/PNG -> (D,H,W) tif stacks
dataloader/         Stack_Loader: one patient = batch
model/              DeepLabv3 (ResNet-18 + ASPP)
train_source.py     source training (CE + Dice)
methods/ODES.py     pruning, acquisition, BN update
methods/freq_shift.py  NEW: FFT / DWT shift score
FFT_weight_2.py     diversity map (missing upstream,
                    re-implemented from the paper)
demo_ODES.py        online loop + Dice / IoU eval
run_all.sh          21-run experiment sweep
make_report.py      tables, plots, LaTeX numbers
```

Upstream repo had no training script, no checkpoint and a missing module; we added these and fixed CPU/GPU and a stale-variable bug.

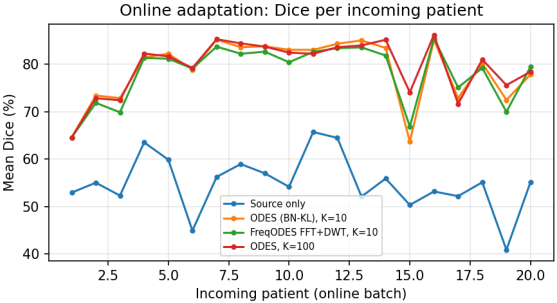
Table 2: Training / adaptation settings

| | |
|---|---|
| Framework | PyTorch 2, timm |
| Source optimiser | Adam, lr $10^{-3}$, wd $10^{-4}$, cosine |
| Batch size / epochs | 8 / 25 |
| Loss | CE + Dice |
| Augmentation | flips, $90^{\circ}$ rot., intensity jitter |
| Adaptation optimiser | Adam, lr $10^{-2}$, 1 step / patient |
| Updated params | BN $\gamma, \beta$ only |
| Budget | $b = 1\%$ px, $K \in \{10, 50, 100\}\%$ |
| Hardware | NVIDIA Tesla T4 (Colab) |
| Seed | 1337 (all runs) |

# Training log and curves




Source training loss (25 epochs, 647 IP slices): $1.07 \rightarrow 0.16$.




Online adaptation: mean Dice of each incoming target patient.

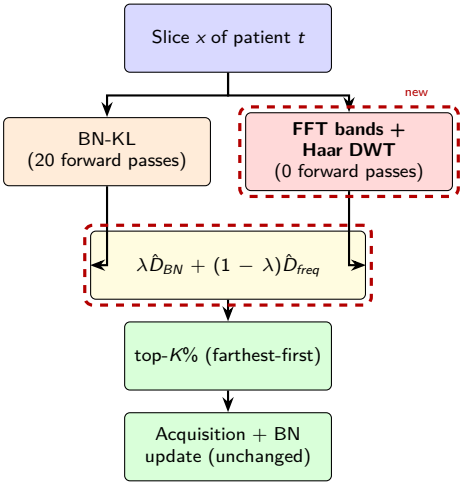
Expert-guided BN updates lift every patient above source-only.

# What we changed: frequency-domain image pruning

**Hypothesis**: MR sequence / scanner shift is directly visible as a change in spatial-frequency energy, so it can be detected from the image itself, without the network.

- **FFT descriptor**: log fraction of energy in 16 concentric radial bands of $|\mathcal{F}(x)|^2$.
- **DWT descriptor**: log relative energy of $LL_3$ and LH/HL/HH sub-bands of a 3-level Haar DWT.
- Reference $\mu_f, \sigma_f$ from 200 random source slices.
- Shift score $D_{freq}(x) = \frac{1}{d}\sum_f \left(\frac{\phi_f(x)-\mu_f}{\sigma_f}\right)^2$
- **Fused**: $D = \lambda \hat{D}_{BN} + (1-\lambda)\hat{D}_{freq}$

Only Stage 3 changes; backbone, acquisition, update and budgets are identical $\Rightarrow$ fair comparison.



# Does the frequency score detect the shift?

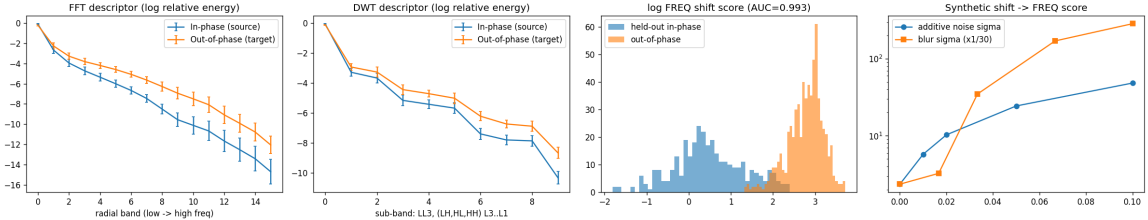



Table 3: Separating held-out IP slices (10 patients) from OOP slices with the shift score alone.

| Descriptor | FFT | DWT | FFT+DWT |
|---|---|---|---|
| ROC-AUC | **0.994** | 0.969 | 0.993 |

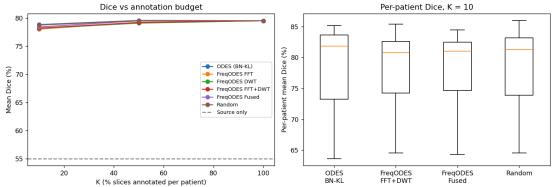
OOP slices carry systematically more high-frequency energy; the score also grows monotonically with synthetic noise and blur.

# FreqODES v1 (frequency score alone) vs. ODES, single seed

Table 4: Same data, model, seed and budget ($b = 1\%$). DSC / IoU in %, time = full online pass over 20 patients on a T4. $p$: paired Wilcoxon vs. ODES at the same $K$.

| | $K = 10\%$ of slices | | | | $K = 50\%$ of slices | | | |
|---|---|---|---|---|---|---|---|---|
| Pruning score | DSC | IoU | $p$ | Time (s) | DSC | IoU | $p$ | Time (s) |
| BN-KL (ODES, replicated) | **78.83** | 66.24 | – | 167 | 79.57 | 67.00 | – | 171 |
| FFT (FreqODES) | 78.02 | 65.03 | 0.005 | 107 | 79.24 | 66.61 | 0.064 | 115 |
| DWT (FreqODES) | 78.15 | 65.28 | 0.027 | 109 | 79.09 | 66.43 | 0.030 | 114 |
| FFT+DWT (FreqODES) | 78.20 | 65.27 | 0.091 | 109 | 79.20 | 66.52 | 0.044 | 114 |
| Fused, $\lambda$=0.5 | 78.40 | 65.45 | 0.159 | 169 | **79.50** | 66.92 | 0.687 | 173 |
| Random | 78.75 | 66.04 | 0.334 | 107 | 79.52 | 67.00 | 1.000 | 113 |

All slices ($K = 100$): DSC 79.52, IoU 66.96  Source only: DSC 54.97, IoU 42.74



# Ablations: pruning $\times$ diversity weighting, and fusion weight

Table 5: Mean DSC at $K = 10$ (cf. Table 3 of [1]).

| Image pruning | D.W. off | D.W. on |
|---|---|---|
| BN-KL (ODES) | 78.73 | 78.83 |
| FFT+DWT (ours) | 78.32 | 78.20 |
| Random | 78.96 | 78.75 |
| Paper (BN / random) | 75.88 / 74.32 | 76.80 / 75.41 |

Table 6: Fusion weight $\lambda$ ($K = 10$).

| $\lambda$ | Mean DSC |
|---|---|
| 0 (pure frequency) | 78.20 |
| 0.25 | 79.09 |
| 0.5 | 78.40 |
| 0.75 | 78.93 |
| 1 (pure BN = ODES) | 78.83 |

Mixing a little BN information into the frequency score ($\lambda = 0.25$) gives the best $K = 10$ result (79.09 vs. 78.83, not significant, $p = 0.75$); diversity weighting changes DSC by $\leq 0.2$ in our runs, much less than reported in the paper.

# Analysis: why v1 alone did not beat ODES (and what it led to)

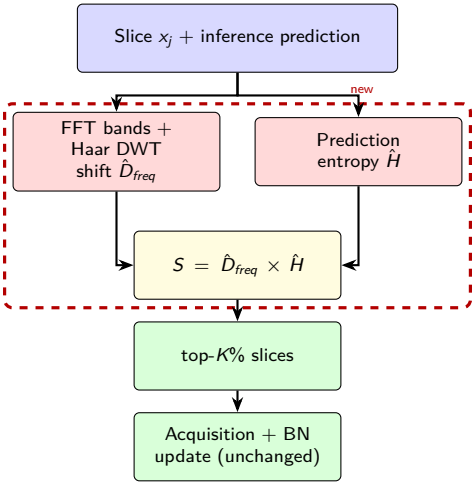
- **Where it helps**: best shift *detector* (AUC 0.994), interpretable, needs no forward passes $\Rightarrow$ full online pass 109 s vs. 167 s ($\approx$ 35% faster). The fused score matches ODES at both budgets (79.50 vs. 79.57 at $K = 50$, $p = 0.687$) and with $\lambda = 0.25$ reaches 79.09 at $K = 10$.
- **Where it hurts**: pure frequency scores trail ODES by 0.6–0.8 DSC at $K = 10$ (FFT: -0.81, $p = 0.005$) and by 0.3–0.5 at $K = 50$.
- **Mismatch between objective and score**: the most frequency-shifted slices are the volume extremes (few or tiny organs, mostly background); labelling them teaches the BN layers little. BN-KL measures what *the network* finds unusual, which correlates better with segmentation error.
- **Small, homogeneous shift**: IP$\rightarrow$OOP shift is global, so nearly every OOP slice is "shifted"; ranking among them is what matters, and random pruning is already strong (78.75, $p = 0.334$).
- **Fix**: a slice is worth labelling if it is *shifted* **and** the model is *unsure* about it $\Rightarrow$ multiply the frequency score by model uncertainty (next slide), and evaluate over 3 seeds.

# FreqODES-U: frequency shift × model uncertainty

**Slice score**

$$S(x_j) = \underbrace{\hat{D}_{freq}(x_j)}_{\text{FFT+DWT shift}} \times \underbrace{\hat{H}(x_j)}_{\text{mean pixel entropy}}$$

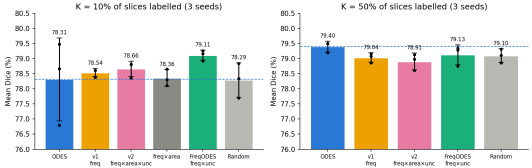
- $\hat{D}_{freq}$: v1 frequency score; $\hat{H}$: mean entropy of the softmax prediction of the slice; each normalised by its batch maximum.
- $\hat{H}$ reuses the inference prediction ODES already makes ⇒ still **0 extra forward passes**.
- Pre-registered design (v2): $\hat{D}_{freq}\times$ organ area $\times\hat{H}$, with two ablations (drop one factor each). The ablation $\hat{D}_{freq} \times \hat{H}$ was best; we report *all* variants.
- Protocol: 3 seeds (1, 2, 12345) × 20 patients; paired Wilcoxon over the 60 patient-runs.



# FreqODES-U results: 3 seeds, same model and budget

Table 7: Mean Dice (%) ± std over 3 seeds; Δ and $p$: paired Wilcoxon vs. ODES over 60 patient-runs; time: full online pass (T4).

| | $K = 10\%$ of slices | | | | $K = 50\%$ of slices | | |
|---|---|---|---|---|---|---|---|
| Selection score | DSC | Δ / $p$ | IoU | Time | DSC | Δ / $p$ | IoU |
| ODES (BN-KL) | 78.31±1.38 | – | 65.51 | 154 s | **79.40**±0.18 | – | 66.90 |
| v1: frequency | 78.54±0.13 | +0.22 / 0.908 | 65.63 | 104 s | 79.04±0.15 | -0.36 / 0.000 | 66.36 |
| v2: freq × area × unc. | 78.66±0.26 | +0.35 / 0.199 | 65.84 | 105 s | 78.91±0.27 | -0.49 / 0.000 | 66.18 |
| Ablation: freq × area | 78.36±0.28 | +0.04 / 0.700 | 65.57 | 104 s | – | – | – |
| **FreqODES-U: freq × unc.** | **79.11**±0.17 | **+0.80 / 0.004** | **66.43** | 104 s | 79.13±0.33 | -0.27 / 0.018 | 66.44 |
| Random | 78.29±0.57 | -0.02 / 0.232 | 65.40 | 102 s | 79.10±0.23 | -0.30 / 0.001 | 66.43 |




At $K = 10\%$ FreqODES-U beats ODES on every seed (78.93, 79.16, 79.26 vs. 76.80, 78.67, 79.48), 8× more stable, ≈33% faster. At $K = 50\%$ ODES stays best.

# Conclusion

- **Replication**: ODES reproduced on CHAOS IP→OOP after completing the public code: 79.52 DSC with all slices annotated (paper 78.61), adaptation gain ≈25 DSC as in the paper.
- **v1 – frequency shift score**: an interpretable FFT/DWT detector of domain shift (AUC 0.994) that needs no forward passes (≈35% cheaper), but on its own it does not beat ODES: a good *shift* detector is not automatically a good *annotation-value* detector.
- **FreqODES-U – frequency × uncertainty**: **79.11 vs. 78.31 DSC at** $K = 10\%$ ($\Delta$+0.80, $p = 0.004$, 3 seeds), about 8× more stable and ≈33% faster; at $K = 50\%$ ODES stays best (79.40 vs. 79.13).
- **Caveats**: best variant of a pre-planned ablation; one dataset pair; 20 patients; 3 seeds.
- **Future work**: CHAOS→DUKE and BMC→RUNMC, more seeds and stream orders, organ-ROI spectra.

# Team contributions

Both members took part in every stage and can explain the whole project; the main responsibilities were:

## Vishnu Vardhan (CB.SC.U4AIE23043)

- Environment, CHAOS download and preprocessing
- Completing the ODES code: training script, missing `FFT_weight_2` module, bug fixes
- Source-model training and ODES replication
- Implementing the FFT/DWT score and FreqODES-U
- GPU experiments on Colab, GitHub repository

## Nithya Sai (CB.SC.U4AIE23021)

- Literature survey and project proposal
- Frequency analysis of IP vs. OOP (spectra, AUC)
- v2 design, ablation plan and 3-seed evaluation
- Statistical analysis (Wilcoxon), figures and results page
- Slides, Elsevier report and viva preparation

# References


[1] M. S. Islam, S. Nag, A. Dutta, S. M. Ahmed, F. F. Niloy, S. Bera, A. K. Roy-Chowdhury. ODES: Online domain adaptation with expert guidance for medical image segmentation. *MICCAI*, 2025.

[2] A. E. Kavur et al. CHAOS challenge – combined (CT-MR) healthy abdominal organ segmentation. *Medical Image Analysis* 69:101950, 2021.

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[4] B. Xie, L. Yuan, S. Li, C. H. Liu, X. Cheng. Towards fewer annotations: active learning via region impurity and prediction uncertainty for domain adaptive semantic segmentation. *CVPR*, 2022.

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[7] S. G. Mallat. A theory for multiresolution signal decomposition: the wavelet representation. *IEEE TPAMI* 11(7):674–693, 1989.

# Thank You

Questions?

Group A12 ▪ Vishnu Vardhan (CB.SC.U4AIE23043) ▪ Nithya Sai (CB.SC.U4AIE23021)

https://github.com/Vishnu868/FreqODES